

Kumppurachchi: K.A.R.R

1. Homogeneous transformation matrix H_1^0 for 3.4.

$$H_1^0 = \begin{bmatrix} 0 & -1 & 0 & 2 \\ 1 & 0 & 0 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

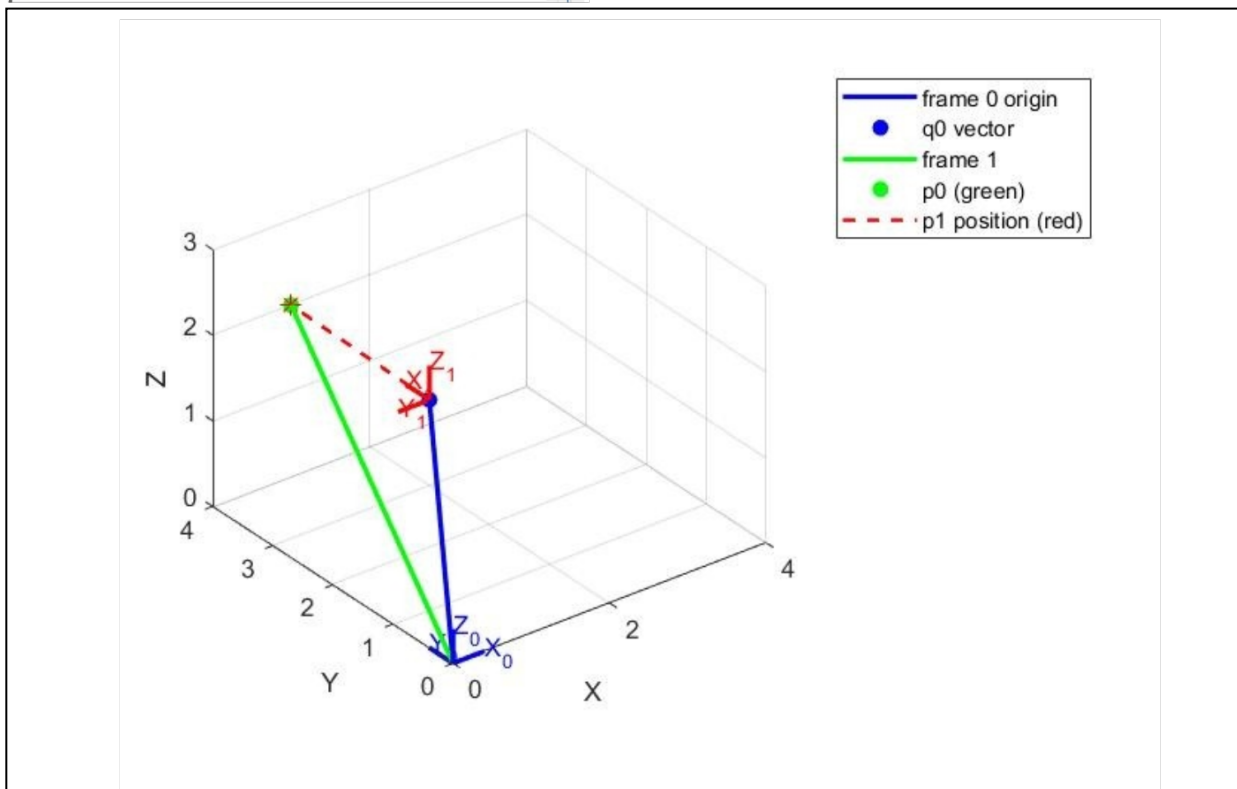
2. MATLAB code for 3.1 ~ 3.6.

```

1 % Lab02 - tasks 3.1 to 3.6
2
3 clc; clear; close all;
4
5 % ---
6 q0_in_0 = [2; 3; 1]; % q0 expressed in frame (0)
7 angle_deg = 90; % rotation about Z to get frame (-9)
8 p_in_1 = [1; 1; 1]; % point p expressed in frame (1)
9 % ---
10
11 % 3.1 Visualize coordinate system (0)
12 figure('Name','Lab02 3.1-3.4');
13 trplot(eye(4), 'frame', '0', 'color', 'b', 'length', 0.4, 'thick', 2);
14 hold on; grid on; axis equal;
15 axis([0 4 0 4 0 3]); xlabel('X'); ylabel('Y'); zlabel('Z');
16
17 % 3.2 Obtain rotation R1_0 and translation t1_0
18 theta = deg2rad(angle_deg);
19 R1_0 = rotz(theta); % rotation matrix 3x3 (frame1 relative to frame0)
20 t1_0 = q0_in_0; % translation vector (origin of (1) in (0))
21
22 % Display R and t
23 disp('R1_0 ='); disp(R1_0);
24 disp('t1_0 ='); disp(t1_0);
25
26 % 3.3 Visualize q0 in the figure using blue color (arrow from origin)
27 % plot a blue arrow from origin to q0_in_0
28 plot3([0 q0_in_0(1)], [0 q0_in_0(2)], [0 q0_in_0(3)], 'b-', 'LineWidth', 2);
29 plot3(q0_in_0(1), q0_in_0(2), q0_in_0(3), 'bo', 'MarkerSize', 4, 'MarkerFaceColor','b');
30
31 % 3.4 Obtain H1_0 and visualize coordinate frame (1) using red color
32 H1_0 = rt2tr(R1_0, t1_0); % homogeneous transform (Robotics Toolbox)
33 % (alternatively: H1_0 = [R1_0 t1_0; 0 0 0 1]);
34 trplot(H1_0, 'frame', '1', 'color', 'r', 'length', 0.4, 'thick', 2);
35
36 % show H1_0 numeric
37 disp('H1_0 ='); disp(H1_0);
38
39 % 3.5 Find p0 and visualize it in the same figure using green color
40 % Convert p1 (in (1)) to homogeneous, then to (0)
41 p1_h = [p_in_1; 1]; % homogeneous coordinates in frame (1)
42 p0_h = H1_0 * p1_h; % p expressed in frame (0)
43 p_in_0 = p0_h(1:3);
44
45 % plot p0 as green arrow from origin
46 plot3([0 p_in_0(1)], [0 p_in_0(2)], [0 p_in_0(3)], 'g-', 'LineWidth', 2);
47 plot3(p_in_0(1), p_in_0(2), p_in_0(3), 'go', 'MarkerFaceColor','g');
48
49 disp('p_in_1 (given) ='); disp(p_in_1);
50 disp('p_in_0 (converted) ='); disp(p_in_0);
51
52 % 3.6 Visualize p1 in the same figure using red color
53 % To plot p1 in the global figure we show its location in (0) coordinates (already computed)
54 plot3([q0_in_0(1) p_in_0(1)], [q0_in_0(2) p_in_0(2)], [q0_in_0(3) p_in_0(3)], 'r--', 'LineWidth', 2);
55 plot3(p_in_0(1), p_in_0(2), p_in_0(3), 'r+', 'MarkerSize', 8);
56
57 legend('frame 0 origin','q0 vector','frame 1','p0 (green)','p1 position (red)', 'Location', 'n', 'hold off');
58
59

```

ons in 3.1 ~ 3.6.



4. Homogeneous transformation matrix H_0^1 for 3.8.

```
H0_1 (inverse transformation) =
    0    1    0   -3
   -1    0    0    2
    0    0    1   -1
    0    0    0    1
```

5. t_0^1 for 3.10.

```
t0_1 (position of frame {0} origin in frame {1}) =
   -3    2   -1
```

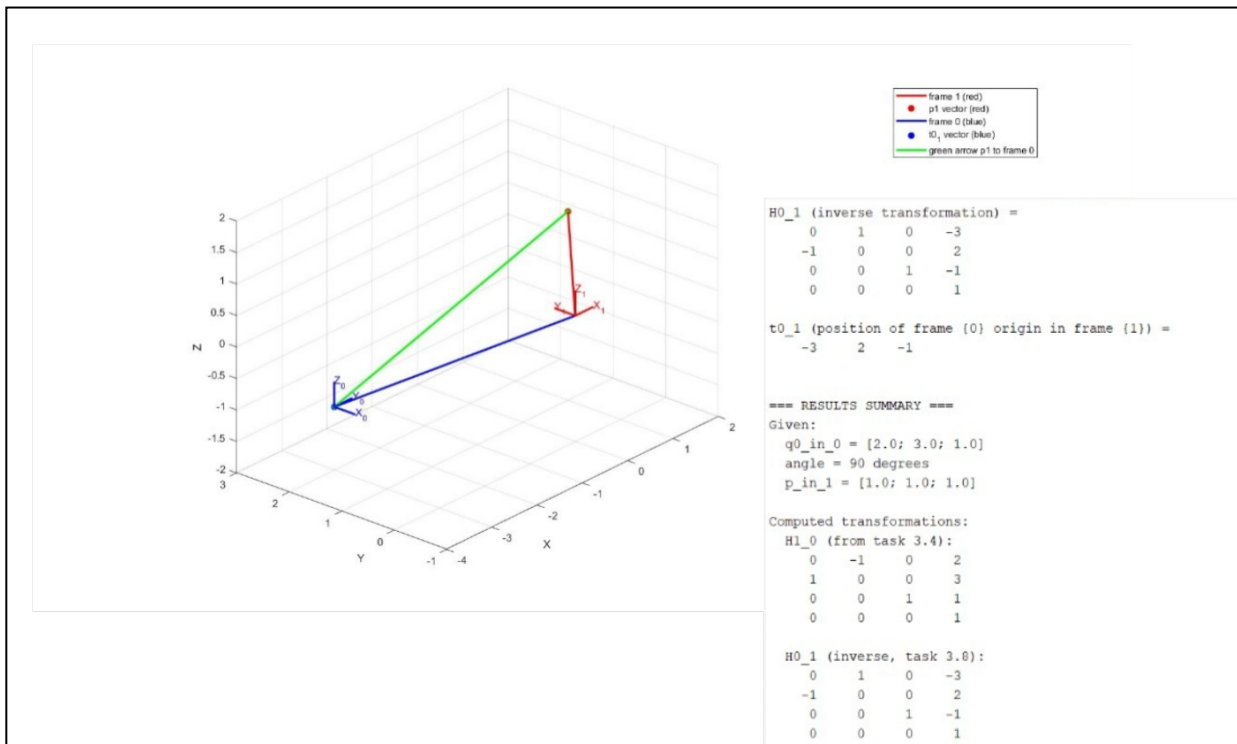
6. MATLAB code for 3.7 ~ 3.11.

```

1 % Lab02 - tasks 3.7 to 3.11
2 % Continuation from tasks 3.1-3.6
3
4 clc; clear; close all;
5
6 % ---
7 % Given values from previous tasks
8 q0_in_0 = [2; 3; 1]; % q0 expressed in frame {0}
9 angle_deg = 90; % rotation about Z to get frame {1}
10 p_in_1 = [1; 1; 1]; % point p expressed in frame {1}
11 % ---
12
13 % First, we need to compute the transformations from previous tasks
14 theta = deg2rad(angle_deg);
15 R1_0 = rots(theta); % rotation matrix 3x3 (frame0 relative to frame1)
16 t1_0 = q0_in_0; % translation vector (origin of {1} in {0})
17 H1_0 = rt2tr(R1_0, t1_0); % homogeneous transform from {0} to {1}
18
19 % 3.7 Use a new MATLAB figure to visualize coordinate system {1} and p1 using red color
20 figure('Name','Lab02 3.7-3.11');
21 trplot(eye(4), 'frame', '1', 'color', 'r', 'length', 0.4, 'thick', 2);
22 hold on; grid on; axis equal;
23 axis([-4 2 -1 3 -2 2]); xlabel('X'); ylabel('Y'); zlabel('Z');
24
25 % Visualize p1 in frame {1} using red color
26 plot3([0 p_in_1(1)], [0 p_in_1(2)], [0 p_in_1(3)], 'r-', 'LineWidth', 2);
27 plot3(p_in_1(1), p_in_1(2), p_in_1(3), 'ro', 'MarkerSize', 6, 'MarkerFaceColor','r');
28
29 % 3.8 Obtain the homogeneous transformation matrix H0_1 (inverse of H1_0)
30
31 % 3.9 Obtain the homogeneous transformation matrix H0_1 (inverse of H1_0)
32 H0_1 = inv(H1_0);
33 % Alternative method using the inverse formula: H0_1 = [R1_0' -R1_0'*t1_0; 0 0 0 1];
34 disp('H0_1 (inverse transformation) =');
35 disp(H0_1);
36
37 % 3.9 Visualize frame {0} with blue color
38 trplot(H0_1, 'frame', '0', 'color', 'b', 'length', 0.4, 'thick', 2);
39
40 % 3.10 Find t0_1 and visualize it on the figure with blue color
41 % t0_1 is the translation part of H0_1, which represents the position of frame {0} origin in frame {1}
42 t0_1 = H0_1(1:3, 4);
43 disp('t0_1 (position of frame {0} origin in frame {1}) =');
44 disp(t0_1);
45
46 % Visualize t0_1 as blue arrow from origin of frame {1}
47 plot3([0 t0_1(1)], [0 t0_1(2)], [0 t0_1(3)], 'b-', 'LineWidth', 2);
48 plot3(t0_1(1), t0_1(2), t0_1(3), 'bo', 'MarkerSize', 6, 'MarkerFaceColor','b');
49
50 % 3.11 Visualize a green arrow from the tip of p1 to the origin of frame {0}
51 % So we draw an arrow from p_in_1 to t0_1
52 plot3([p_in_1(1) t0_1(1)], [p_in_1(2) t0_1(2)], [p_in_1(3) t0_1(3)], 'g-', 'LineWidth', 2);
53 plot3(p_in_1(1), p_in_1(2), p_in_1(3), 'go', 'MarkerSize', 6);
54
55 % Add legend
56 legend('frame 1 (red)', 'p1 vector (red)', 'frame 0 (blue)', 't0_1 vector (blue)', 'green arrow');
57
58 % Display results summary
59 fprintf('\n=== RESULTS SUMMARY ===\n');
60 fprintf('Given:\n');
61 fprintf('  q0_in_0 = [3.1f; 3.1f; 3.1f]\n', q0_in_0);
62 fprintf('  angle = %.0f degrees\n', angle_deg);
63 fprintf('  p_in_1 = [3.1f; 3.1f; 3.1f]\n', p_in_1);
64
65 fprintf('\nComputed transformations:\n');
66 fprintf('  H1_0 (from task 3.4):\n');
67 disp(H1_0);
68 fprintf('  H0_1 (inverse, task 3.8):\n');
69 disp(H0_1);
70 fprintf('  t0_1 = [3.4f; 3.4f; 3.4f]\n', t0_1);
71
72 hold off;

```

7. Final output MATLAB figure for the operations in 3.7 ~ 3.11.



8. Homogeneous transformation table.

Requirement	MATLAB script to satisfy the requirement	Homogeneous transformation matrix result
$O_0X_0Y_0Z_0$ to $O_1X_1Y_1Z_1$	<pre>Script for H1_0: t1_0 = [0.0; 1.0; 1.0]; R1_0 = eye(3); H1_0 = rt2tr(R1_0, t1_0);</pre>	<pre>H1_0 (Frame {0} to Frame {1} - Table surface): 1 0 0 0 0 1 0 1 0 0 1 1 0 0 0 1</pre>
$O_0X_0Y_0Z_0$ to $O_2X_2Y_2Z_2$	<pre>Script for H2_0: t2_0 = [-0.5; 1.5; 1.0]; R2_0 = roty(deg2rad(45)); H2_0 = rt2tr(R2_0, t2_0);</pre>	<pre>H2_0 (Frame {0} to Frame {2} - Box location): 1.0000 0 0 -0.5000 0 1.0000 0 1.5000 0 0 1.0000 1.0000 0 0 0 1.0000</pre>
$O_0X_0Y_0Z_0$ to $O_3X_3Y_3Z_3$	<pre>Script for H3_0: t3_0 = [-0.5; 1.5; 3.0]; R3_0 = rotx(deg2rad(180)) * rotz(deg2rad(90)); H3_0 = rt2tr(R3_0, t3_0);</pre>	<pre>H3_0 (Frame {0} to Frame {3} - Camera location): 0 1.0000 0 -0.5000 1.0000 0 0 1.5000 0 0 -1.0000 3.0000 0 0 0 1.0000</pre>

